

Kevin (Jai-Hua) Yen

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EDUCATION

The State University of New York at Buffalo
Doctor of Philosophy in Biostatistics

New York, USA
Aug. 2024 - Present

National Taiwan University
Master of Science in Biometry

Taipei, Taiwan
Sep. 2017 - Aug. 2019

National Tsing Hua University
Bachelor of Science in Quantitative Finance

Hsinchu, Taiwan
Sep. 2013 - Jun. 2017

SKILLS

- **Programming:** Python, R, SQL, Golang, Matlab, SAS, C, C++
- **Tools:** Linux, Docker, Git, LaTeX

INDUSTRY EXPERIENCE

QNAP
Senior AI Software Engineer
AI Software Engineer

Taipei, Taiwan
Apr. 2024 - Jun. 2024
Feb. 2022 - Apr. 2024

- Trained a ResNet50 image classification model with a 70% F1-Score across 250 image classes and optical character recognition (OCR) models using vision transformers with 90% accuracy to enable efficient customer photo searches
- Leveraged CLIP (Contrastive Language-Image Pre-Training), a multimodal model, to integrate semantic search into the NAS search engine, empowering customers to search for photos and files using human-readable descriptions instead of conventional keywords
- Deployed models on edge devices, including customer NAS systems with Tensor Processing Unit (TPU) and Neural Processing Unit (NPU) acceleration, improving data security and inference speed within constrained computing resources
- Mentored junior personnel on programming, project organization, and training OCR models

TutorABC
AI Algorithm Engineer

Taipei, Taiwan
Jun. 2020 - Feb. 2022

- Created an 81% accurate customer churn detection model using random forest to decrease churn rates by 5%
- Developed an XGBoost customer purchase detection model with a 78% accuracy rate; utilized explainable machine learning method (SHAP value) to boost purchase rates and formulate targeted sales strategies
- Extracted key customer feedback using computer vision (YOLO, OCR) and natural language processing (TFIDF, BERT) models, enhancing data analysis for additional product features and improved user experience
- Accelerated the extraction, transformation, and loading (ETL) processes by 50%; consolidated by implementing Airflow for machine learning operations (MLOps)
- Mentored a junior engineer in programming, project structuring, and MLOps

RESEARCH EXPERIENCE

National Taiwan University

Graduate Student Research Assistant

Taipei, Taiwan

Sep. 2017 - Aug. 2019

- Derived an adjusted Chao2 species richness estimator, reducing underestimation by 15% and mitigating a 10% overestimation in scenarios with very low corrected denominators while mitigating identity errors common in species surveys
- Conducted data cleaning on weed and plant cover species surveys conducted in Soft Bridge County, Taiwan, and Grand St. Bernard Pass, Switzerland

TEACHING EXPERIENCE

The State University of New York at Buffalo

Teaching Assistant

New York, USA

Aug. 2024 - Now

National Taiwan University & Taipei Medical University

Teaching Assistant Conference Lecturer

Taipei, Taiwan

Feb. 2019

National Taiwan University

Teaching Assistant

Taipei, Taiwan

Sep. 2017 - Jan. 2019

PUBLICATION

Yen, J.-H., Chiu, C.-H. (2020). Richness estimation with species identity error. *Proceedings 62nd International Statistical Institute World Statistics Congress*, **Volume 6**, 401-408.

Yen, J.-H., Chiu, C.-H. (2019, August 18-23). *Richness estimation with species identity error* [Poster presentation]. The 62nd International Statistical Institute World Statistics Congress, Kuala Lumpur, Malaysia.

Yen, J.-H., Chiu, C.-H. (2018, July 22-27). *Richness estimation in the existence of species identity error* [Poster presentation]. The 61st Symposium of the International Association for Vegetation Science, Bozeman, Montana, USA.